- **Post secondary / career prospects in the field** (post secondary programs, experiential learning opportunities, career options and trends etc)
    - Kai

George Brown school of Mechanical Engineering Technology

- Access to lots of technical knowledge and hands on experience to work with various technological systems, training in machine workshops, labs and technology like wind turbines, electric Vespas (zero emission version of motor scooter), and mainly robotics [1]

Mechatronics Engineering - University of Waterloo

- Focus is on merging mechanical hardware with software development

- Students use MATLAB/Simulink (math simulator dashboard used to map out physic formulas and simulate how a machine will behave before building it) [2]
- These students complete up to two years of paid co-op work. At companies like Magna International, one of the world's largest companies in Canada for automotives, students write python, C++, MATLAB test scripts to check electric vehicle drivetrains (group of parts in a vehicle that delivers power from engine or motor to wheels) monitoring battery voltage stability across high stress endurance cycles [3]

Robotics Engineering - University of Toronto

- Focus is on system designs focusing on artificial intelligence and autonomous machine behavior
- Usually use low-level programming languages like C and advanced robotic hardware environments
- Through the co-op, undergraduate spend 12 to 16 months at firms like BlackBerry QNX where they help test software microkernels (minimalist operating system design that only manages necessary things like CPU scheduling, basic memory management, etc) inside QNX Neutrino operating system, which runs active driver-safety assistance screens inside over 235 million car globally [4]

Electrical and Computer Engineering - McGill University

- Focus on designing, programming, and manufacturing electronic microchips and hardware components
- Students use VHDL/Verilog languages (not for app writing but designing physical hardware pathways inside a computer chip) inside Xilinx Vivado (a software engine used to flash code and test custom microchips) development software to build custom configurations for the chips [5]
- Obviously, the students program microcontroller chips where they write safety loops that interpret real time data from vehicle accident sensors, ensuring the chip triggers safety deployment signals in every short times [6]

Career Options

- **Characterized by high salaries, excellent job security and versatility**

Autonomous system engineer, smart infrastructure specialist, embedded firmware developer, biomedical engineer, process automation engineer, etc

Key Terms:

Entry level, post-grad

Remember NOT co-op or internships (including internal and external in high school and university)

Some Sources:

https://joinhandshake.com/blog/students/top-10-jobs-for-computer-science-majors/

Detailed descriptions:

Biomedical engineer:

- Use of biological material to create a lot of things that benefit society including pharmaceuticals, nutritional supplements, environmentally friendly chemicals, and biofuels(combustible fuel derived from organic matter)
- Lots of healthcare and energy related technology involved such as creating detection methods, artificial kidney dialysis, or biosensors used for diseases diagnostics, health monitoring, and overall health safety
- Bio technology leads to to careers that overall work in government, private companies, or clinical laboratories to produce and develop technologies that help society [7]

Autonomous system engineer:

- These engineers write automated navigation logic for delivery drones, factory rovers, and self-driving vehicles
- Languages commonly used: Python, C++, ROS2
- Example: An engineer working at a robotics firm like Clearpath Robotics programs automated warehouse utility carts. They use LiDAR hardware (a

sensor that shoots out millions of tiny laser beams per second to map a room in 3d so self-driving machines can see) to map structural pillars inside a warehouse. They use computer science to ensure vehicles can reroute their navigation paths in real time. [8]

Smart Infrastructure Specialist:

- Specialists upgrade traditional civil engineering structures like bridges, tunnels, dams by pairing them with live computer monitoring systems
- They use tech like Autodesk Tandem/Bentley iTwin(Specialized 3D modeling programs used to build digital exact virtual clones of real infrastructure) to host the 3d Visual dashboard
- Example: Bentley's iTwin IoT platform has been deployed to monitor dam performance at California's New Bullards Bar Dam, providing real time visibility into structural safety, which is the same approach used on overpasses, where sensor data can alert inspectors to potential issues weeks before physical damage forms [9]

Trends

- Artificial intelligence will be in high demand for engineers in designing and developing the software, hardware, and interfaces power AI-based products
- Global investment in generative AI engineering climbed from 4.38 billion in 2025 to 5.01 billion in 2026. [10] India is leading global adoption acceleration with a projected 23.8% compound annual growth rate through the decade. China follows at 23% driven by commercial electric vehicle programs and then the United States at 21.5% driven by domestic defense procurement updates. [11]
- Automotives OEMS like Toyota and Honda started to transition to generative software engines. The algorithm they use embeds constraints like aerodynamic drag, ride height, etc directly into design generation process, allowing engineers to optimize vehicle structures without traditional back and forth between design and engineering teams. [12]
- One in two businesses experience a successful cyberattack in the past year [13], and as industries begin to create IoT devices and smart infrastructure that can potentially be vulnerable to these attacks, with an estimated calculation of up to 15.63 trillion by 2029 from 2018. [14], many engineers will become an increasingly important role in creating technologies to prevent these attacks. Civil engineers will need to improve their ability to secure smart city grids, electrical engineers will have to find ways to protect power distribution networks from hacks
- According to the International Federation of Robotics(IFR), South Korea has kept the world's highest robot density, at an average annual rate of 7% since 2019 to reach around 1220 per 10000 employees. [15] Singapore ranks second at 818,

and China, despite a massive human workforce, has accelerated its domestic robotic employment by 17% year on year, having around 54% of all new industrial robot installations worldwide. [15] This trend demonstrates the growing shift from human operated workforces to testing rapidly the idea of robots that can replace what humans can do.

Learning Opportunities:

At Louisiana Tech University in first year engineering, students use an arduino microcontroller, dc wheel motors, a breadboard, jumper wires, and ultrasonic distance sensors to physically assemble a mobile robot chassis, wire dc motors, and plug distance sensor chips into a breadboard. THey use computer science via C++ and flash the code onto the microcontroller via USB. Specifically for this university, students drop a car into a physical maze.[16]

At UC Berkeley, students can take a course where they create a single pixel camera in their first year. They build a physical circuit by plugging a photoresistor and a resistor to a breadboard and connect that to a hardware microcontroller board to use for their computer. and use python to run an automated image scanning process. They write scripts that coordinate a physical projector to shine light arrays across an object while the python code calculates the light reflected to the photoresistors.[17]

Extra:

According to the FBI's 2025 Internet Crime Report, released April 6, 2026, cybercrime losses to Americans crossed $20.877 billion in 2025 — a 26% increase from the $16.6 billion reported in 2024. It was also the first year in IC3's 25-year history that the complaint center received more than one million complaints in a single year, averaging nearly 3,000 complaints per day throughout 2025. [18] Cybersecurity Ventures independently projects total global cybercrime costs at approximately $10.5 trillion for 2026 [19] The IBM Cost of a Data Breach Report 2025 found that the global average cost of a single data breach fell to $4.44 million in 2025 — the first decline in five years, down 9% from $4.88 million in 2024 — with IBM attributing most of that reduction to faster detection and containment driven specifically by security AI and automation tools, highlighting the growing role of engineering-driven solutions in cybersecurity [20]

https://iclg.com/news/23737-us-cybercrime-up-by-more-than-a-quarter-in-just-one-year/
https://cybersecurityventures.com/official-cybercrime-report-2025/
https://www.bakerdonelson.com/webfiles/Publications/20250822_Cost-of-a-Data-Breach-Report-2025.pdf

The Cengage 2025 Graduate Employability Report found that only 51% of graduates believed they had sufficient AI skills for the jobs they applied to. Only 30% of 2025

graduates secured full-time work in their field, with nearly half feeling unprepared for entry-level roles — underscoring a widening gap between what engineering programs currently teach and what employers now expect. [21]
https://www.forbes.com/sites/michaeltnietzel/2025/09/10/new-report-as-skills-gap-grows-job-market-for-college-grads-at-5-year-low/

According to IEEE Spectrum's Top Programming Languages 2025 (published September 23, 2025) — which combines data from search engines, Stack Overflow, GitHub repositories, and job postings — Python ranked #1 overall across all three of IEEE's composite rankings: Spectrum (active use among engineers), Jobs (employer demand), and Trending (current growth). Java ranked #2, and C++ remained in the top 5 specifically due to its dominance in AI runtimes and robotics. IEEE's analysis notes that the number of Stack Exchange questions posted per week dropped to just 22% of 2024 levels as engineers shift toward using AI assistants like Cursor and Claude for coding help instead of public forums — signaling a fundamental shift in how engineers interact with programming tools.
https://spectrum.ieee.org/top-programming-languages-2025

Based on 365 Data Science's analysis of 903 AI engineer job postings from Glassdoor US (updated April 23, 2026): Python appears in 71% of postings, AWS in 32.9%, Azure in 26%, and Java in 22%. Only 2.5% of positions target entry-level candidates, meaning prior hands-on experience through internships or projects is essential.
Based on their separate analysis of 1,000 ML engineer job postings from Indeed US (updated April 23, 2026): Python leads at 56.3%, SQL at 26.1%, Java at 21.1%. For deep learning frameworks, PyTorch leads at 39.8% of postings and TensorFlow at 37.5%. The average salary for ML engineers in 2026 reached $166,000 — a $35,000 increase from 2025.
https://365datascience.com/career-advice/career-guides/ai-engineer-job-outlook-2025/
https://365datascience.com/career-advice/career-guides/machine-learning-engineer-skills/
AI-related job postings grew 7.5% in the past year while total job postings across all fields fell 11.3%, according to PwC's 2025 Global AI Jobs Barometer. Workers with AI skills now command a 56% wage premium over equivalent non-AI positions.
https://scrimba.com/articles/best-ai-engineering-courses-and-certifications-in-2026/

IEEE Innovation at Work offers a direct post-secondary learning pathway for engineers through its eLearning library, with confirmed course tracks including:
AI & Machine Learning industry track
Cybersecurity & Digital Trust track
Semiconductors track

IEEE | Rutgers Online Mini-MBA: Artificial Intelligence — a professional development program co-developed with Rutgers University
Courses award CEU (Continuing Education Unit) and PDH (Professional Development Hour) credits, recognized for professional engineering licensure. The platform was confirmed live as of August 2025.
https://innovationatwork.ieee.org/elearning/
https://innovationatwork.ieee.org/elearning/

For engineers seeking to build Python and AI skills at the post-secondary level, IEEE Innovation at Work offers a dedicated AI & Machine Learning industry track with self-paced and live virtual courses covering Python-based machine learning, generative AI, natural language processing, and AI integration for engineering workflows. The platform also offers the IEEE | Rutgers Online Mini-MBA: Artificial Intelligence — a professional development program co-developed with Rutgers University designed for engineers moving into AI leadership roles. All courses award CEU and PDH credits recognized for professional engineering licensure
https://innovationatwork.ieee.org/individuals/ai-ml/
The IEEE | Rutgers Online Mini-MBA: Artificial Intelligence is a joint program co-developed by IEEE and Rutgers Business School faculty, designed specifically to help engineers and technical professionals understand and apply AI tools — including Python-driven machine learning and data analytics — in real engineering and business contexts. The program is completed in 12 weeks or less through a self-paced, fully online format pairing expert-led video instruction with interactive assessments, live office hours, and a capstone project. Confirmed course modules include: Introduction to AI, Data Analytics, Process Optimization, AI for Supply Chain, AI in Finance, Ethics in AI, and AI's Impacts on the Future of Work: New Careers, Coworkers, and Competencies.
https://innovationatwork.ieee.org/professional-development/rutgers-online-mini-mba-artificial-intelligence/

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